

PDC Reconstruction Literature Survey and SAMURAI Terminal Air-MS Gap Analysis

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Contents

1	TL;DR	3
2	The problem: air multiple scattering at the SAMURAI terminal	3
2.1	Geometry of the proton flight path	3
2.2	Measured σ values	4
2.3	Gap to design specifications	4
2.4	Why this matters for the science floor	5
3	PDC reconstruction reference frame	5
3.1	Hardware foundations	5
3.2	Algorithm spine	7
3.3	Performance achieved across SAMURAI campaigns	7
4	Gap analysis	7
4.1	Performance: σ values	7
4.2	Methodology	7
4.3	Material budget and vacuum configuration	7
4.4	Experimental configuration and geometry	7
5	Broader MINOS+SEASTAR chain context	7
5.1	SEASTAR programme timeline	7
5.2	Companion detectors	7
5.3	Reviews and resources	7
5.4	Caveats from the broader bibliography	7
6	Implications and next steps	7
6.1	Short-term data acquisition action items	7
6.2	Medium-term simulation and analysis directions	7
6.3	Open questions	7
6.4	Non-goals	7
7	References	7
7.1	Hardware and methodology (peer-reviewed, foundational)	7
7.2	Algorithm references	7
7.3	PhD theses with PDC methodology	7
7.4	Physics papers using PDC1/PDC2 (confirmed/implicit)	7
7.5	Excluded / context-only	7
7.6	Reviews and topical collections	7

7.7	Construction proposals and internal notes	7
7.8	Project-internal references	7

1 TL;DR

This document surveys the published reference frame for the SAMURAI PDC1/PDC2 drift-chamber chain and contrasts it with our current Geant4 reconstruction performance, in order to localise the air multiple-scattering gap that drives our PDC2 spatial resolution one order of magnitude above design.

- **Reference frame.** The PDC1/PDC2 chain has no dedicated peer-reviewed publication. The canonical references are the SAMURAI overview paper by Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013) §3.4¹, the RIKEN internal technical note *Detector-PDC.pdf*², and the Kahlbow PhD thesis (TU Darmstadt, 2019)³ which documents an independent reconstruction of the same hardware.
- **Design specifications.** Combining Kobayashi 2013 §3.4¹ with the SAMURAI 2012 construction proposal⁴, the PDC1/PDC2 chain targets 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution ($\Delta p/p$). Commissioning RI-beam runs reached $\Delta p/p \approx 1/1500$ ¹; the original proton-mode design value is $\Delta p/p \approx 1/700$ as quoted in the RIBF-TAC05 (2005) construction proposal⁵.
- **Our measured σ_y (PDC2).** In our Geant4 + reconstruction chain we observe σ_y (PDC2) = 12.83–14.93 mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration⁶. The realistic-air values sit roughly an order of magnitude away from the 1 mm hardware design specification¹.
- **Air MS dominates and the configuration choice matters.** A quadrature decomposition⁶ attributes the all-air budget to $\sigma_A \approx 11.19$ mm from Region A (cavity + beam pipe, ~ 4.3 m) and $\sigma_B \approx 7.43$ mm from Region B (Exit Window $\rightarrow$ PDCs, ~ 2 m). The stage D inputs from the same memo⁷ give σ_y (PDC2) = 7.49 mm if Region A is vacuum and σ_y (PDC2) = 13.50 mm if both Region A and Region B are air. Whether Region A is air or vacuum during the experiment is therefore a configuration question that selects between these two values; this document does not assert which one corresponds to the actual run.

See §6 for proposed action items.

2 The problem: air multiple scattering at the SAMURAI terminal

2.1 Geometry of the proton flight path

The deuteron target sits at world coordinates $(-12.488, 0.0009, -1069.458)$ mm, which corresponds to yoke-local coordinates $(-545.5, 0, -919.9)$ mm under the -30° yoke rotation about Y ⁸. Because the SAMURAI dipole cavity (`vchamber_box`) extends over $z \in [-1750, +1750]$ mm

¹T. Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013), DOI: [10.1016/j.nimb.2013.05.089](https://doi.org/10.1016/j.nimb.2013.05.089).

²RIKEN Nishina Center, *Detector-PDC* technical note, <https://www.nishina.riken.jp/ribf/SAMURAI/image/Detector-PDC.pdf>.

³J. Kahlbow, PhD thesis, TU Darmstadt (2019), TUpriints 9192, <https://tuprints.ulb.tu-darmstadt.de/9192/>.

⁴SAMURAI Collaboration, *SAMURAI Construction Proposal* (2012), https://ribf.riken.jp/SAMURAI/120425_SAMURAIConstProp.pdf.

⁵SAMURAI Collaboration, *RIBF-TAC05 SAMURAI construction proposal* (2005), https://ribf.riken.jp/RIBF-TAC05/10_SAMURAI.pdf.

⁶See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

⁷See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §6.

⁸See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §1.1–§1.3 for coordinates, yoke rotation, and Region A/B definitions.

in yoke-local frame, the target at $z = -919.9$ mm is *physically inside the dipole cavity* rather than upstream of it. Two regions therefore separate the target from the PDCs. **Region A** (≈ 4.3 m) covers the in-cavity flight from the target through the cavity exit (2.67 m) and onwards through the downstream beam-pipe (`VacuumDownstream`, ≈ 1.6 m) up to the Exit Window; this region is controlled by the Geant4 macro switch `/samurai/geometry/BeamLineVacuum`. **Region B** (≈ 2.0 m) covers free flight in the world volume from the Exit Window at world (170.5, 0, 3187.5) mm to PDC1 at (700, 0, 4000) mm (0.97 m) and on to PDC2 (1.00 m), gated by `/samurai/geometry/FillAir`. Inside Region A the proton bends $\sim 30^\circ$ in the 1.15 T field, but the arc-length vs. straight-line difference stays below 10%⁸.

2.2 Measured σ values

Table 1 reproduces the robust half-width $\sigma = (p_{84} - p_{16})/2$ measured at PDC1 and PDC2 across three vacuum/air configurations and three truth momenta, with $|\vec{p}| = 627$ MeV/c protons.

Condition	(p_x, p_y) [MeV/c]	N	$\sigma_x(\text{P1})$	$\sigma_y(\text{P1})$	$\sigma_x(\text{P2})$	$\sigma_y(\text{P2})$	σ_{θ_x}	σ_{θ_y}
all_air	(0, 0)	50	3.071	9.736	3.669	12.826	12.279	9.240
all_air	(100, 0)	50	3.838	10.492	5.377	12.751	15.768	8.277
all_air	(0, 20)	50	3.372	13.039	4.065	14.927	21.709	7.846
beamline_vacuum	(0, 0)	50	1.528	4.061	2.086	6.801	11.834	5.910
beamline_vacuum	(100, 0)	50	2.067	4.888	3.060	8.211	8.100	5.786
beamline_vacuum	(0, 20)	49	1.667	4.789	2.363	7.459	14.289	5.421
all_vacuum	(0, 0)	50	0.147	0.515	0.447	0.947	6.798	2.558
all_vacuum	(100, 0)	50	0.154	0.364	0.416	0.999	4.075	1.956
all_vacuum	(0, 20)	50	0.174	0.411	0.396	1.002	6.913	2.299

Table 1: σ values from Geant4 + reconstruction ensemble; σ in mm, σ_θ in mrad. Robust half-width $(p_{84} - p_{16})/2$. Source: docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md §4.

2.3 Gap to design specifications

The realistic-air spatial $\sigma_y(\text{PDC2})$ sits roughly an order of magnitude away from the 1 mm hardware design figure, and only the all-vacuum ablation reaches the design floor. Table 2 summarises the comparison; angular and momentum entries follow the same structure.

Quantity	Design / commissioning	Our current state	Source
Position σ at PDC	1 mm	~ 13 mm (all-air) / ~ 7.5 mm (A=vac) / ~ 1.0 mm (all-vac)	Kobayashi 2013 §3.4 ¹ + memo §4 ⁶
Angular σ	2 mrad	5.4–9.2 mrad (σ_{θ_y} , all-air & <code>beamline_vacuum</code>) / 2.0–2.6 mrad (all-vacuum)	Kobayashi 2013 §3.4 ¹ + memo §4 ⁶
RI commissioning $\Delta p/p$	1/1500	not measured in this ablation	Kobayashi 2013 ¹
Proton-mode design $\Delta p/p$	$\sim 1/700$	not measured in this ablation	RIBF-TAC05 (2005) ⁵

Table 2: PDC reconstruction performance: design / commissioning targets vs. our current Geant4 + reconstruction state. Position and angular rows compare against the Kobayashi 2013 §3.4 hardware specification; the two $\Delta p/p$ rows are not yet propagated end-to-end in our ablation, so we have $\sigma_y(\text{PDC2})$ but no single $\Delta p/p$ figure to report.

2.4 Why this matters for the science floor

The PDC2 σ_y entries above are not the end of the story: they propagate backwards through the Runge–Kutta tracker and the SAMURAI magnetic field map to a target-momentum uncertainty $\sigma_{p,\text{target}}$, which is the reconstruction science floor for any experiment in which the target sits inside the magnet and the outgoing proton is measured downstream (target $\rightarrow$ magnet $\rightarrow$ PDC). This document does not re-derive $\sigma_{p,\text{target}}$ propagation values; the per-method numerical breakdown lives in the RK error-analysis deep-dive memo⁹.

3 PDC reconstruction reference frame

3.1 Hardware foundations

The PDC1/PDC2 hardware specification is documented across four canonical sources: Kobayashi 2013 §3.4¹, the RIKEN internal `Detector-PDC.pdf` tech note², the SAMURAI 2012 construction proposal⁴, and the older RIBF-TAC05 (2005) construction proposal⁵. The proton-arm magnet is the SAMURAI superconducting H-shape dipole described by Sato et al., providing a 7 Tm bending power, a maximum field of 3 T, an 80 cm pole gap and a 2 m pole face, with the field map computed in OPERA-3d / TOSCA¹⁰. The drift cells follow the Walenta geometry with an 8 mm drift distance, arranged as five planes (U/V/X/U/V) at $\pm 45^\circ/0^\circ$ over a $1700 \times 800 \text{ mm}^2$ active area, filled with Ar + 25% iC_4H_{10} (or alternatively Ar + 50% C_2H_6) and read out via ~ 810 channels². The chambers sit at $\sim 59^\circ$ on the proton arm with the design targets of 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution, with commissioning RI runs reaching $\Delta p/p \approx 1/1500$ ¹ and a proton-mode design specification of $\Delta p/p \approx 1/700$ ⁵. The signal chain feeds an ASD shaper into AMT-VME TDCs at 0.78 ns/ch⁵, where the AMT chip is the TDC ASIC originally described by Arai¹¹. The boundary between the dipole-cavity vacuum (Region A) and the downstream air volume (Region B) is the SAMURAI Exit Window described by Shimizu et al.: a Kevlar foil of 280 μm backed by a Mylar foil of 75 μm ¹². The matching offline reconstruction pipeline is the RIKEN ROOT-based drift-chamber analysis described by Otsu et al.¹³. Together these four canonical sources cover the hardware geometry and design specifications, but none of them publishes a head-to-head proton-arm reconstruction performance paper; that gap is the motivation for §3.2 (algorithm spine) and §3.3 (performance achieved).

⁹See `docs/reports/reconstruction/momentum_reco/rk/rk_error_analysis_deep_dive_20260426_zh.pdf` for $\sigma_{p,\text{target}}$ propagation under MC, Profile, Laplace, MCMC, and MC-PDC.

¹⁰H. Sato et al., IEEE Trans. Appl. Supercond. **23**, 4500308 (2013), DOI: [10.1109/TASC.2012.2237225](https://doi.org/10.1109/TASC.2012.2237225).

¹¹Y. Arai, Nucl. Instr. Meth. A **453**, 365 (2000), DOI: [10.1016/S0168-9002\(00\)00658-6](https://doi.org/10.1016/S0168-9002(00)00658-6).

¹²Y. Shimizu et al., Nucl. Instr. Meth. B **317**, 739–742 (2013), DOI: [10.1016/j.nimb.2013.08.059](https://doi.org/10.1016/j.nimb.2013.08.059).

¹³H. Otsu et al., Nucl. Instr. Meth. B **376**, 175 (2016), DOI: [10.1016/j.nimb.2016.02.056](https://doi.org/10.1016/j.nimb.2016.02.056).

Item	Value	Source
PDC location	$\sim 59^\circ$ on proton arm	Kobayashi 2013 §3.4
Active area	$1700 \times 800 \text{ mm}^2$	Detector-PDC tech note
Drift cell	Walenta, 8 mm drift	Detector-PDC tech note
Plane configuration	U/V/X/U/V at $\pm 45^\circ/0^\circ$	Detector-PDC tech note
Primary gas	Ar + 25% $i\text{C}_4\text{H}_{10}$	Detector-PDC tech note
Alternative gas	Ar + 50% C_2H_6	Detector-PDC tech note
Channels	~ 810	Detector-PDC tech note
Readout TDC	AMT-VME, 0.78 ns/ch	RIBF-TAC05 2005 + Arai 2000
Magnet type	superconducting H-dipole	Sato 2013
Bending power	7 Tm	Sato 2013
Max field	3 T	Sato 2013
Gap	80 cm	Sato 2013
Pole face	2 m	Sato 2013
Field map source	OPERA-3d / TOSCA	Sato 2013
ExitWindow	Kevlar 280 μm + Mylar 75 μm	Shimizu 2013
Design position σ	1 mm	Kobayashi 2013 §3.4
Design angle σ	2 mrad	Kobayashi 2013 §3.4
Design $\Delta p/p$	0.8%	Kobayashi 2013 §3.4
Commissioning RI $\Delta p/p$	1/1500	Kobayashi 2013
Proton mode $\Delta p/p$ design	$\sim 1/700$	RIBF-TAC05 2005

Table 3: PDC1/PDC2 + magnet + ExitWindow hardware specifications. Sources cited inline; full bibliographic entries in §7.

- 3.2 Algorithm spine
- 3.3 Performance achieved across SAMURAI campaigns
- 4 Gap analysis
 - 4.1 Performance: σ values
 - 4.2 Methodology
 - 4.3 Material budget and vacuum configuration
 - 4.4 Experimental configuration and geometry
- 5 Broader MINOS+SEASTAR chain context
 - 5.1 SEASTAR programme timeline
 - 5.2 Companion detectors
 - 5.3 Reviews and resources
 - 5.4 Caveats from the broader bibliography
- 6 Implications and next steps
 - 6.1 Short-term data acquisition action items
 - 6.2 Medium-term simulation and analysis directions
 - 6.3 Open questions
 - 6.4 Non-goals
- 7 References
 - 7.1 Hardware and methodology (peer-reviewed, foundational)
 - 7.2 Algorithm references
 - 7.3 PhD theses with PDC methodology
 - 7.4 Physics papers using PDC1/PDC2 (confirmed/implicit)
 - 7.5 Excluded / context-only
 - 7.6 Reviews and topical collections
 - 7.7 Construction proposals and internal notes
 - 7.8 Project-internal references